YAKEEN

NEET 2.0 2027

Physics | Lecture - 3

Motion in a Plane

By- Manish Raj (MR Sir)



Todays Goal

H/w of lecture-2 ✓

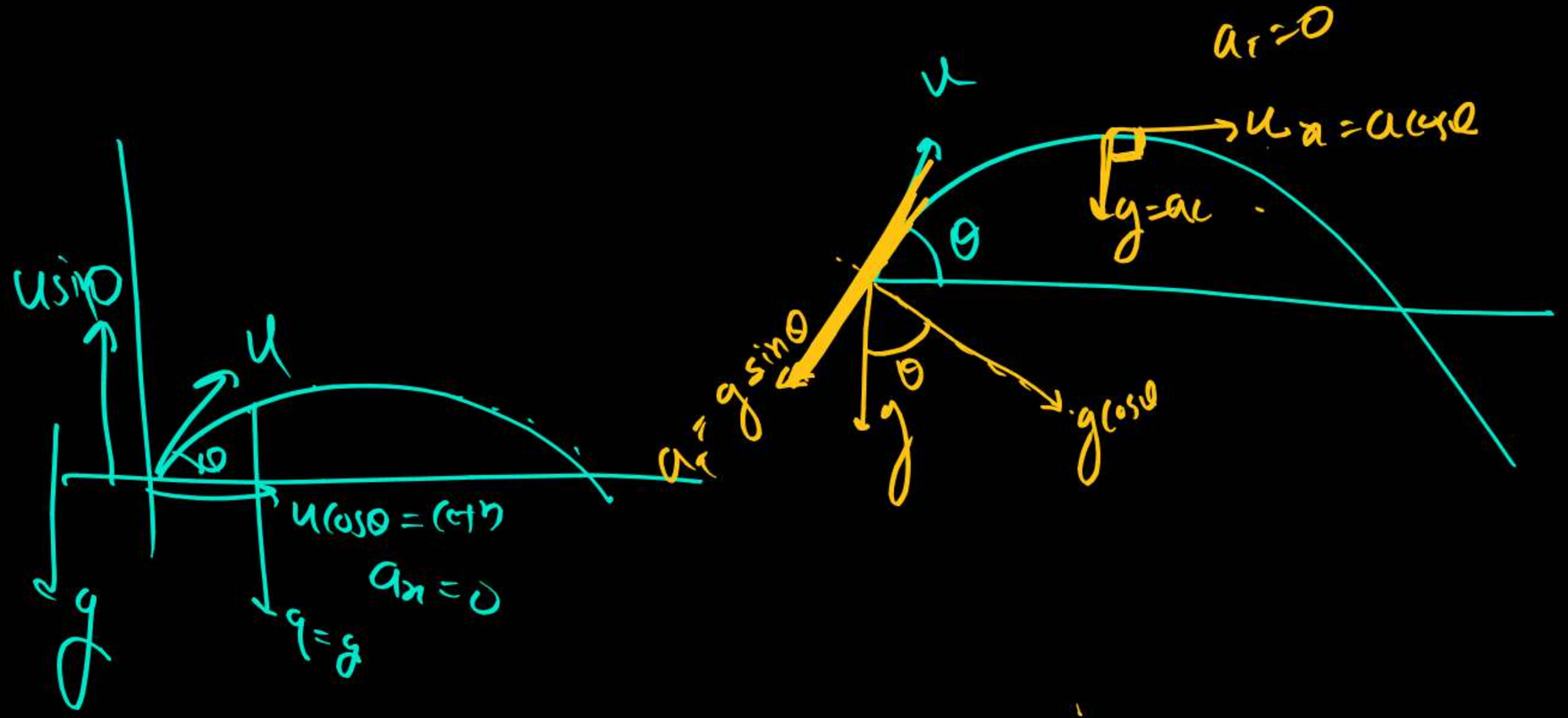
General 2-D motion ; Horizontal Projectile motion. ✓

→ 9mp NEET *

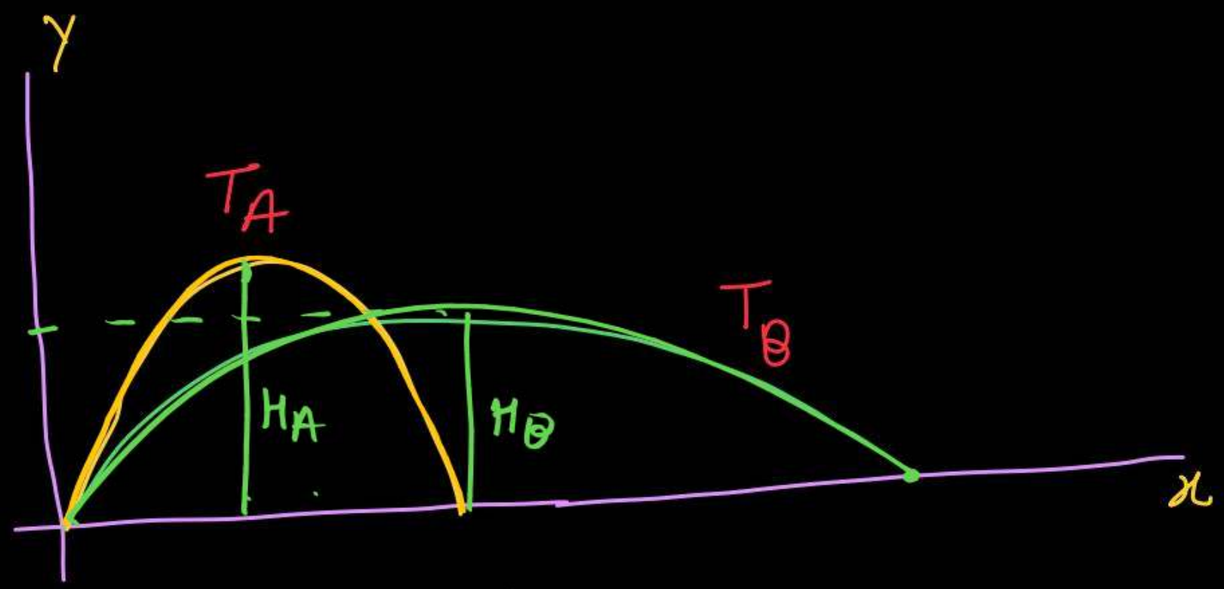
(1) Which of the following physical quantity remains constant in projectile Motion.

(a) speed ✗ (b) velocity ✗ (c) ✓ acceleration (d) ✗ Horizontal velocity

(e) ✗ tangential acceleration.



(2)



(a) ✗ $T_A = T_B$

(b) ✓ $T_A > T_B$

(c) ✗ $T_A < T_B$

Which of the following is correct :-

$$T_f = \frac{2u_y}{g} = \frac{2u\sin\theta}{g}$$

$$H_{max} = \frac{u_y^2}{2g} = \frac{(u\sin\theta)^2}{2g}$$

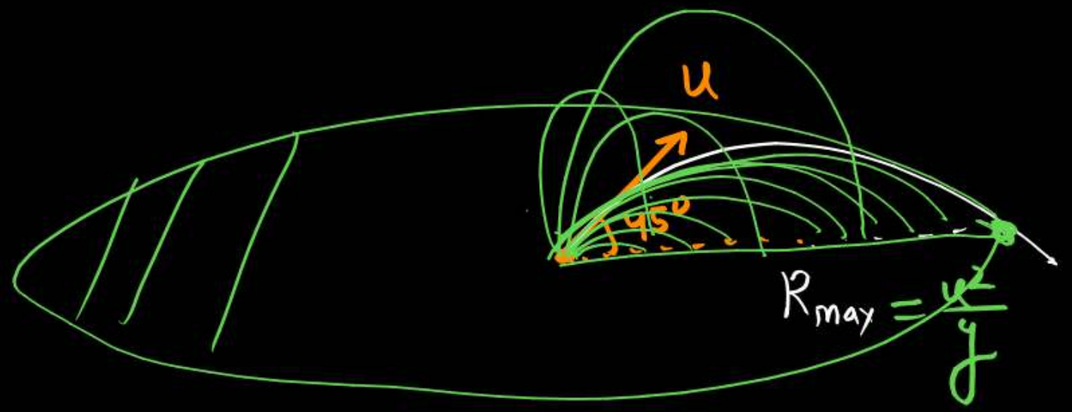
H_{max} large

$H_{max} \propto u_y^2$

$T_f \propto u_y$

3

Ball is Projected with u at diff^n Angle many times then find maximum Area can be covered by Projectile.



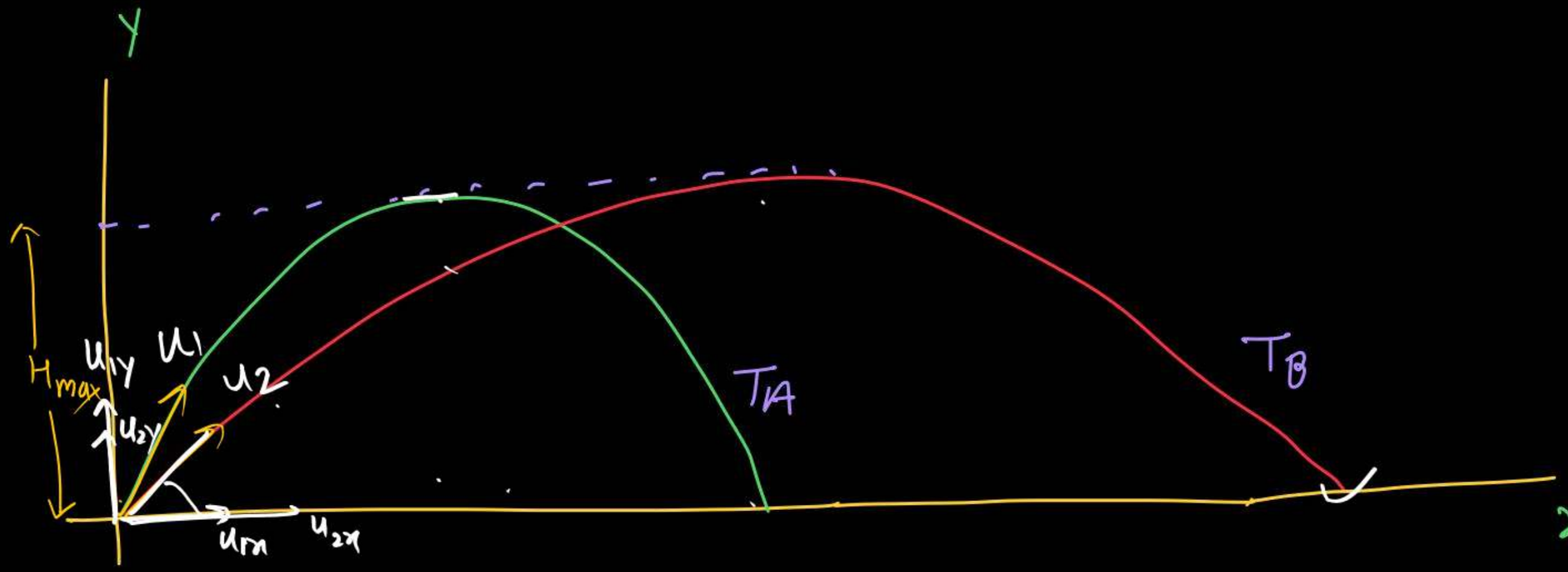
$$R_{max} = \frac{u^2(\sin 2\theta)}{g} = \frac{u^2}{g}$$

$$\text{Area}_{max} = \pi(\text{Radius})^2$$

$$= \pi(R_{max})^2$$

$$= \pi\left(\frac{u^2}{g}\right)^2$$

(4)



Compair u_1 & u_2

$T_A = T_B$

$T_A > T_B$

$T_A < T_B$

$T_f = \frac{2u_y}{g}$

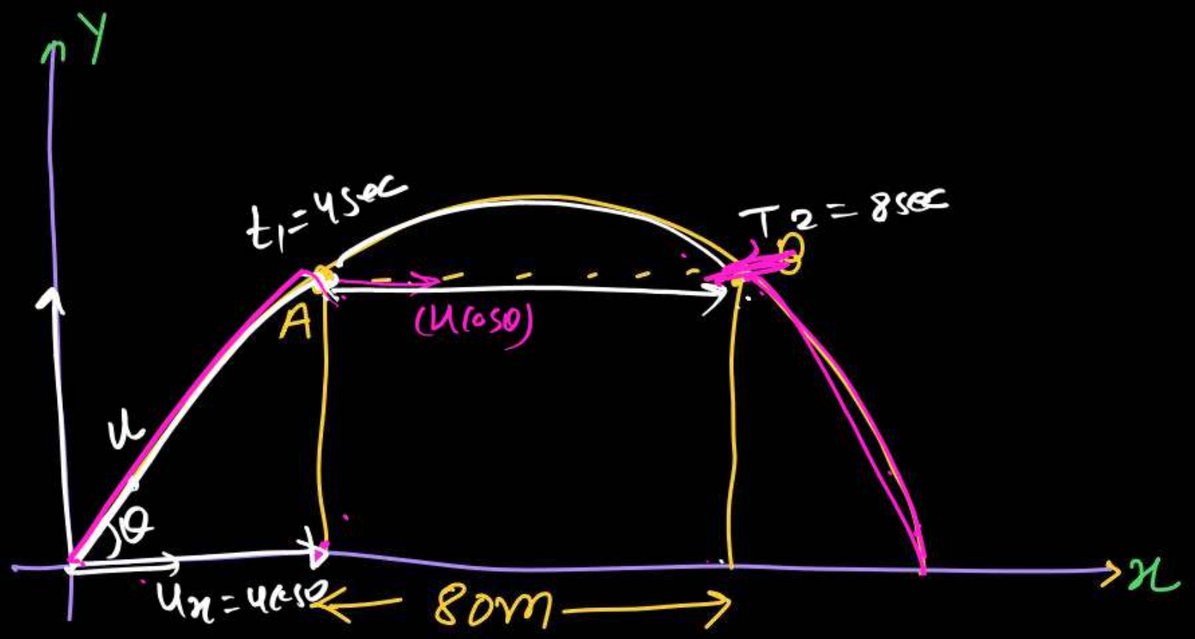
$H_{max} = \frac{u_y^2}{2g}$ (same)

$R_A = u_{x_A} T_f$ $\quad R_B = u_{x_B} T_f$

* Horizontal velocity of B is greater

* $(u_1)_A < (u_2)_B$

(5)



Ball is at same height
$t_A = 4$sec & $t_B = 8$sec then
find speed of Projection
& Angle of Projection.

$$\frac{u\sin\theta = 60}{u\cos\theta = 20}$$
$$\tan\theta = 3$$

Horizontal motion from A to B

$$x_{AB} = u_x t$$
$$80 = u\cos\theta \times 4$$
$$u_x = u\cos\theta = 20 \quad -(1)$$

$$t_A + t_B = T_f$$
$$8 + 4 = 12\text{sec} = \frac{2u\sin\theta}{g}$$

$$u_y = 60 = u\sin\theta \quad -(11)$$

$$\vec{u} = u_x\hat{i} + u_y\hat{j}$$
$$= 20\hat{i} + 60\hat{j}$$
$$|u| = \sqrt{(20)^2 + (60)^2}$$

6/ Ball is projected at two diffrent angle such that range are same & time of flight $t_1 = 4sec$ & $t_2 = 6sec$. then find Horizontal Range. (IIT)

α = 1st Angle of Projection

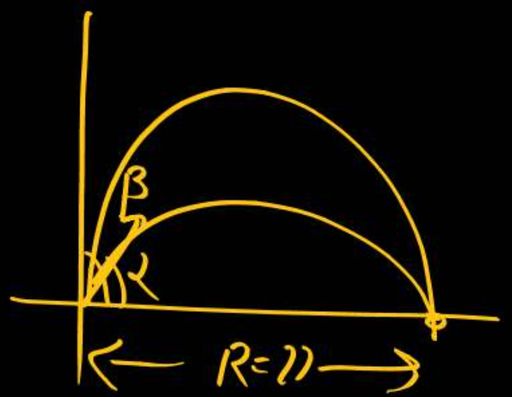
β = 2nd " " "

$$R = \frac{u^2 \sin(2\alpha)}{g} = \frac{u^2 2\sin\alpha\cos\alpha}{g}$$

$$\alpha + \beta = 90 \quad -(1)$$

$$T_1 = \frac{2u\sin\alpha}{g}$$

$$t_2 = \frac{2u\sin\beta}{g}$$



(1) × (2)

$$t_1 t_2 = \frac{4u^2 \sin\alpha \sin(90-\alpha)}{g}$$

$$4\times 6 = \frac{4u^2 \sin\alpha\cos\alpha}{g} = \frac{2R}{}$$

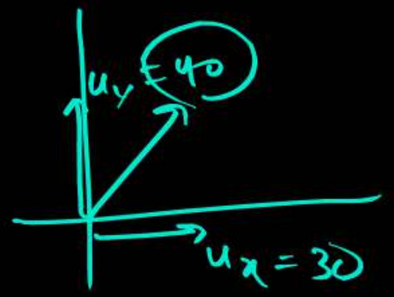
$$2 \times 6 = R$$

$$R = 12m$$

7 Ball is Projected and its position at time t is $\vec{r} = 30t\,\hat{i} + (40t - 5t^2)\,\hat{j}$

Then find Range :-

Solution



$$\vec{r} = 30t\,\hat{i} + (40t - 5t^2)\,\hat{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = 30\,\hat{i} + (40 - 5\times 2t)\,\hat{j}$$

Velocity at t: $\vec{v}_t = 30\,\hat{i} + (40 - 10t)\,\hat{j}$

$$\vec{u}_{t=0} = 30\,\hat{i} + 40\,\hat{j}$$

$$T_f = \frac{2\times 4}{g} = 8\text{ s}$$

$$H_m = 80\text{ m}$$

$$R = 30\times 8 = 240\text{ m}$$

(8) Ball is Projected and its position at time t is $\vec{r} = 30t\,\hat{i} + (40t - 4t^2)\,\hat{j}$

Then find Range :-

Solution

$$\vec{r} = 30t\,i + (40t - 4t^2)\,\hat{j}$$

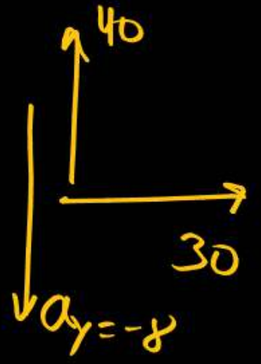
$$\vec{v} = 30i + (40 - 8t)\,\hat{j}$$

$$\vec{V}_{t=0} = 30i + 40\hat{j}$$

$$\vec{a} = 0 + 0 - 8 \times 1\,\hat{j}$$

$$\vec{a} = -8\hat{j}$$

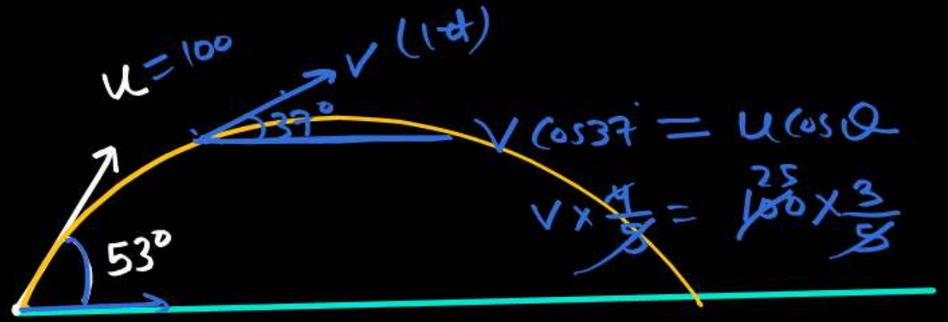
$a_x = 0$



$$T_f = \frac{2u_y}{g} = \frac{2 \times 40}{8} = 10\,sec$$

MR Scan (+1)

Ball 9 is projected with 100m/s at 53° then find its speed when it is moving at 37° from Horizontal. (H|W)



$V\cos 37 = u\cos\theta$

$V \times \frac{4}{5} = 100 \times \frac{3}{5}$

$V = 75\,m/s$

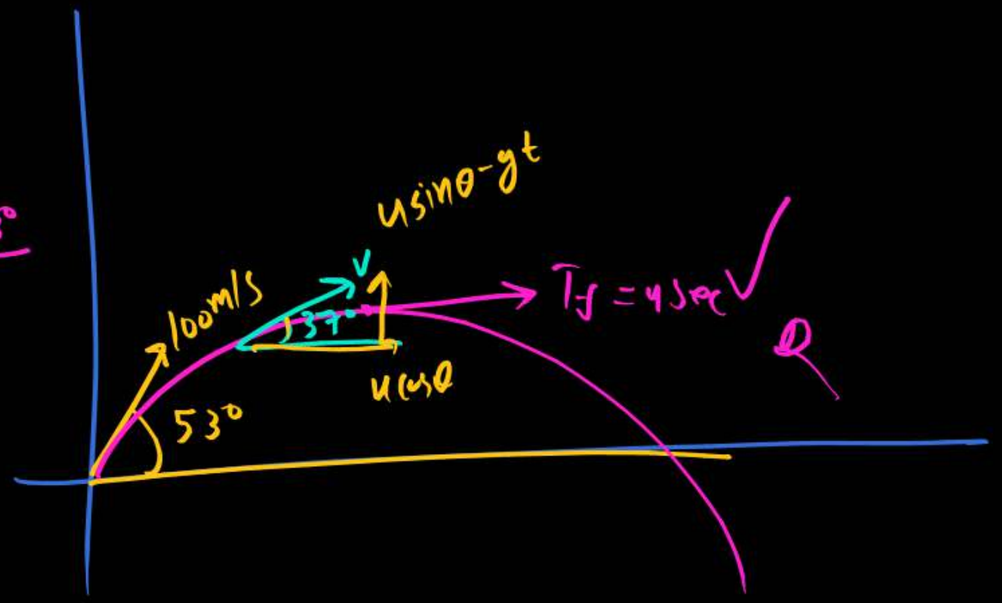
(Q) (10) Ball is projected with speed 100m/s at 53° then find time when its velocity parallel to x-axis and velocity 37° from x axis.

$$T_f = \frac{2u\sin\theta}{g}$$

$$= \frac{2\times 100\times \sin 53^\circ}{10}$$

$$= \frac{4}{\cancel{10}} \times \frac{\cancel{8}}{\cancel{5}}$$

$$= 8 \text{ sec}$$

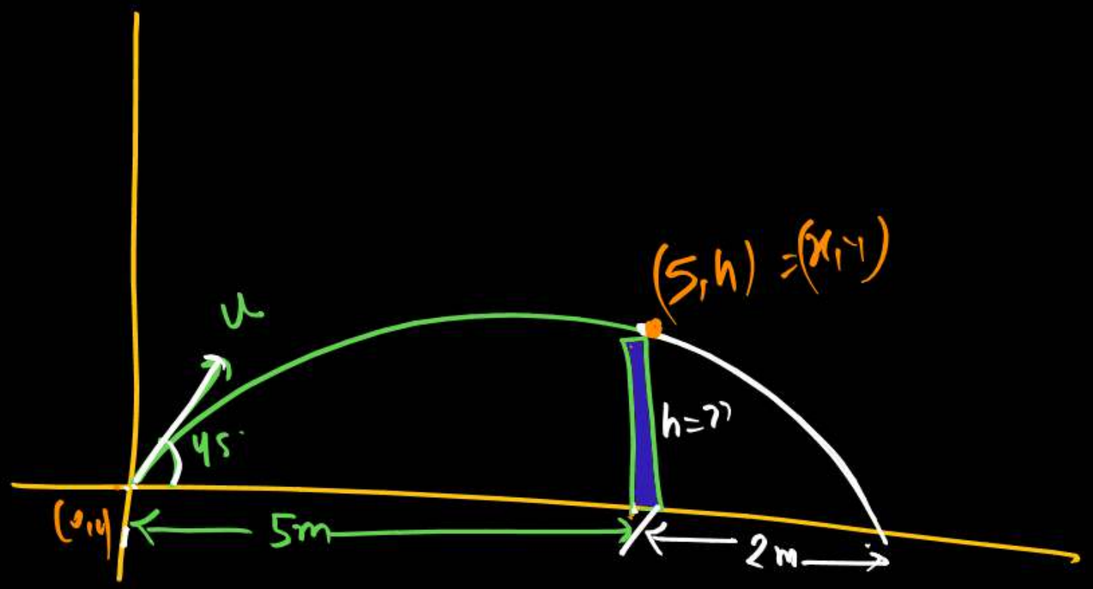


$$\tan\alpha = \frac{u\sin\theta - gt}{u\cos\theta}$$

$$\tan 37^\circ = \frac{100\sin 53^\circ - 10t}{100\cos 53^\circ}$$

Find t

(11) Ball is projected at angle 45° and it touches top of Pole at distⁿ 5m and fall other side 2m from Pole then find height of Pole.



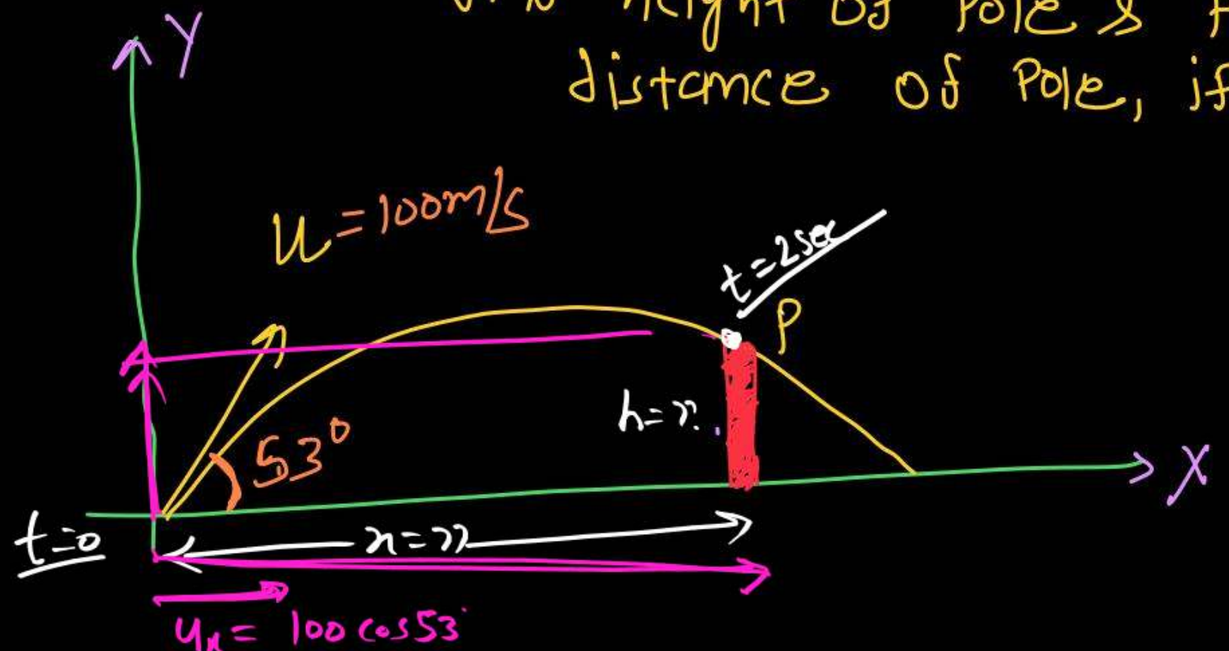
$$y = x\tan\theta\left(1-\frac{x}{R}\right)$$

$$h = 5\tan 45^\circ\left(1-\frac{5}{7}\right)$$

$$h = 5\times 1\left(\frac{7-5}{7}\right)$$

$$= \frac{10}{7}\text{ m}$$

(12) find height of Pole & Horizontal distance of Pole, if Ball touches the Pole at $t = 2\,sec$



$$x = u_x t + \frac{1}{2} a_x t^2$$

$$x = 60 \times 2 = 120\,m \checkmark$$

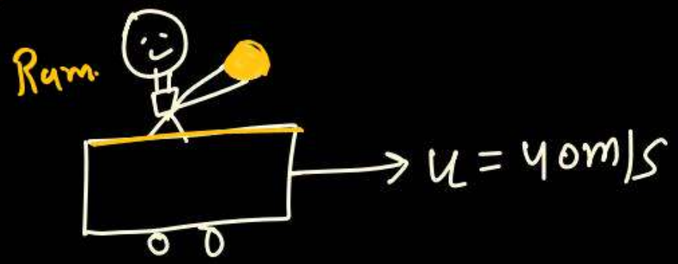
$$y = u_y t + \frac{1}{2} a_y t^2$$

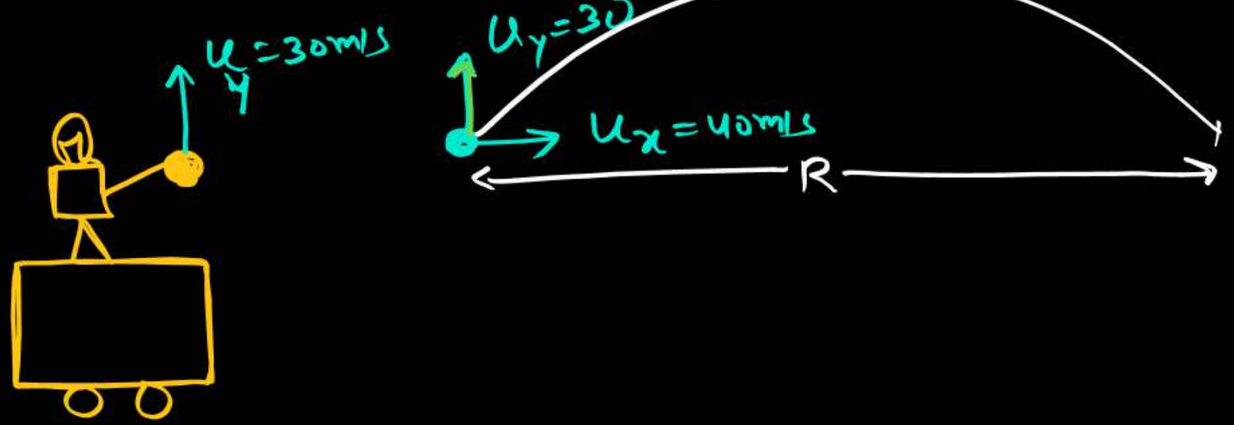
$$h = 80 \times 2 - \frac{1}{2} \times 10 \times (2)^2$$

$$= 160 - 20$$

$$= 140\,m$$

(13) If Ramlal Project Ball vertically up with velocity 30 m/s, then from a bus find H_{max} & Range of Ball





$$T_f = \frac{2u_y}{g} = 6s$$

$$R = 40 \times 6 = 240m$$

$$H_R = 45m$$

Q. If eqn of Trajectory $y = \sqrt{3}x - \frac{x^2}{\sqrt{3}}$ then find Angle of Projection & Range

$$y = \sqrt{3}x - \frac{x^2}{\sqrt{3}}$$

$$y = \sqrt{3}x - \frac{x^2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$y = \sqrt{3}x\left[1 - \frac{x}{3}\right]$$

Compare with $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

$$\tan\theta = \sqrt{3}$$

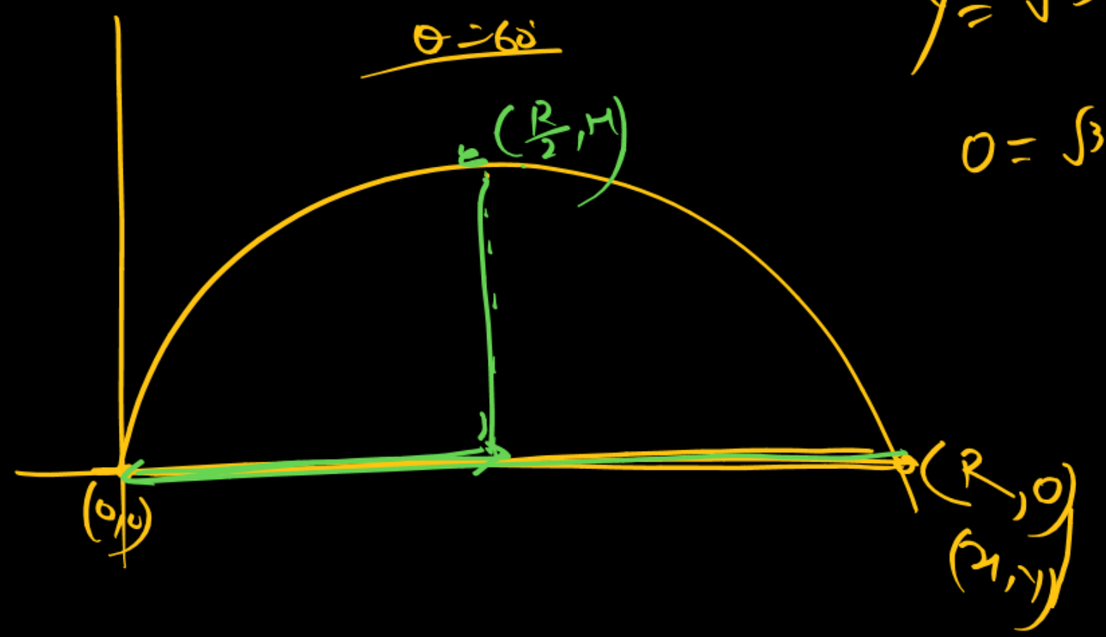
$$\theta = 60^\circ$$

$$R = 3m$$

Q gf eqn of Trajecto $y = \sqrt{3}x - \frac{x^2}{\sqrt{3}}$ then find Angu of Projecta & Range

$\tan\theta = \sqrt{3}$

$\theta = 60°$



$y = x\tan\theta\left(1 - \frac{x}{R}\right)$

$H = \frac{R}{2}\tan\theta\left[1 - \frac{R}{2R}\right]$

$H = \frac{R}{2}\tan\theta\left[\frac{1}{2}\right]$

$$H = \frac{R\tan\theta}{4}$$

$y = \sqrt{3}x - \frac{x^2}{\sqrt{3}}$

$0 = \sqrt{3}R - \frac{R^2}{\sqrt{3}}$

$\frac{R}{\sqrt{3}} = \frac{R^2}{\sqrt{3}}$

$R = 3m$

(15) Ball is Projected with speed 100 m/s at angle 37° then find time when Ball is moving ⊥r to intial dirⁿ of Projection.

$\sin 37° = \frac{3}{5}$

$$t = \frac{u}{g\sin\theta}$$

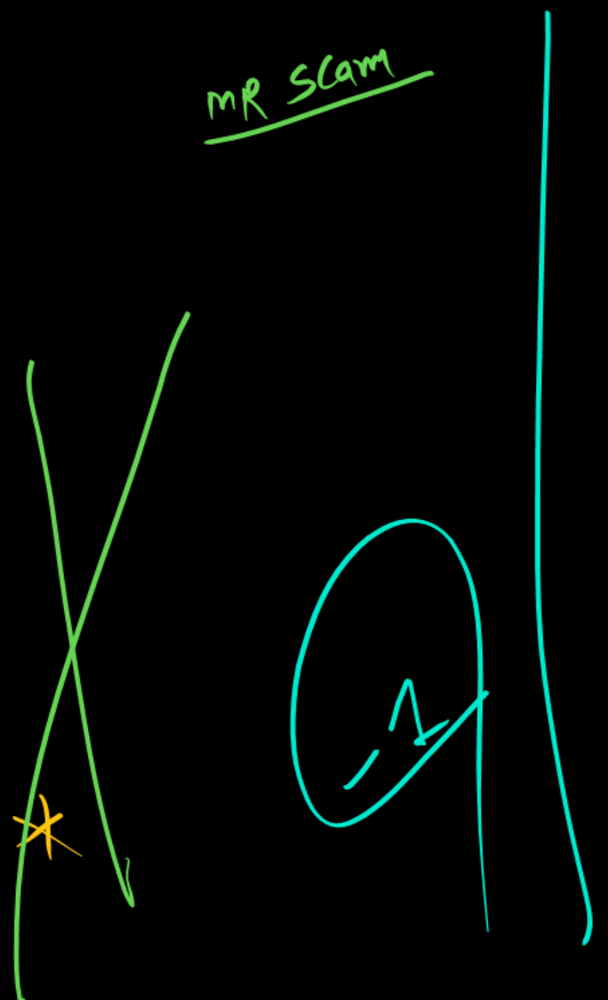
$$= \frac{100}{10 \times \frac{3}{5}}$$

$$t = \frac{100}{6}$$

$$t \simeq 16.6 \text{ sec}$$

MP Scam

Not Possible



Copy in Note

① to ⑮

Question 16

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For angles of projection of a projectile $(45^\circ - \theta)$ and $(45^\circ + \theta)$, the horizontal ranges described by the projectile are in the ratio of: -

AIPMT 2006 (+4 /-1)

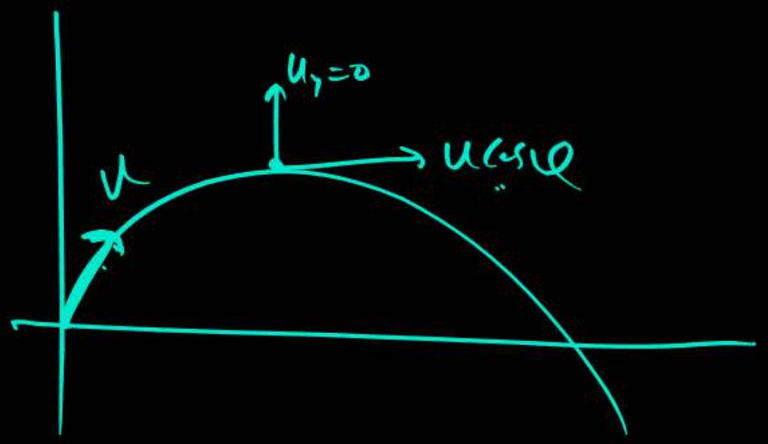
1. 1 : 1 ✓
2. 2 : 3
3. 1 : 2
4. 2 : 1

Question 13

The speed of a projectile at its maximum height is half of its initial speed. The angle of projection is: -

AIPMT (Pre) 2010 (+4 /-1)

1. 15°
2. 30°
3. 45°
4. 60°



$u\cos\theta = \frac{u}{2}$

$\cos\theta = \frac{1}{2}$

$\theta = 60°$

Question

A missile is fired for maximum range with an initial velocity of 20 m/s.

If $g = 10\ m/s^2$, the range of the missile is: - **AIPMT (Pre) 2011 (+4 /-1)**

1. 40 m
2. 50 m
3. 60 m
4. 20 m

$R_{max} = \frac{u^2}{g}$

Question

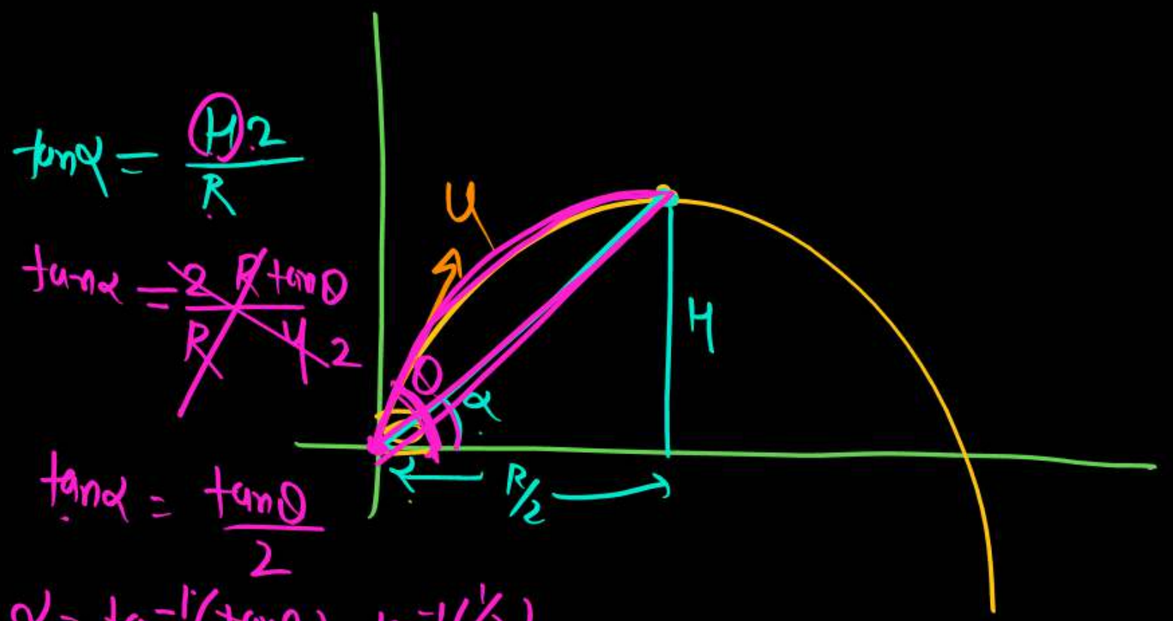
(19) copy

A projectile is fired at an angle of 45° with the horizontal. Elevation angle of the projectile at its highest point as seen from the point of projection, is:

AIPMT (Mains) 2011 (+4 /-1)

1. 45°
2. 60°
3. $\tan^{-1}\left(\frac{1}{2}\right)$
4. $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$H = \frac{R \tan\theta}{4}$



$\tan\theta = \frac{H \cdot 2}{R}$

$\tan\alpha = \frac{2}{R} \frac{R \tan\theta}{4 \cdot 2}$

$\tan\alpha = \frac{\tan\theta}{2}$

$\alpha = \tan^{-1}\left(\frac{\tan\theta}{2}\right) = \tan^{-1}(1/2)$

Question

20

The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile is: -

AIPMT (Pre) 2012 (+4 /-1)

1. $\theta = \tan^{-1}(2)$
2. $\theta = 45°$
3. $\theta = \tan^{-1}\left(\frac{1}{4}\right)$
4. $\theta = \tan^{-1}(4)$

$H = \frac{R}{4}\tan\theta$

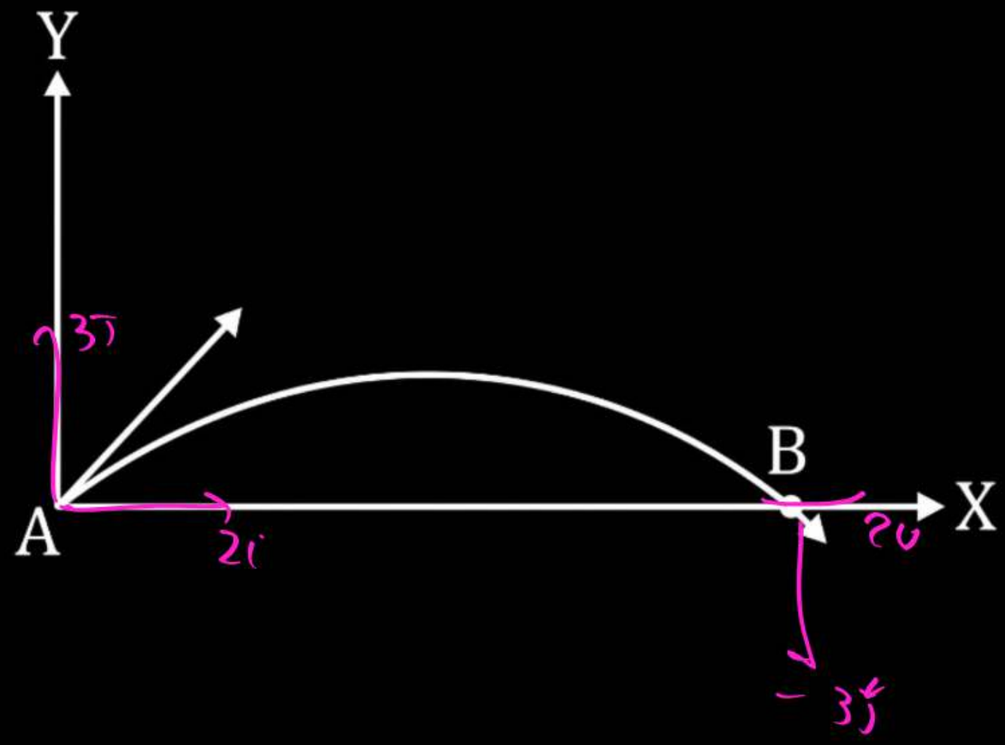
$\tan\theta = 4$

Question (21)

The velocity of a projectile at the initial point A is $(2\hat{i} + 3\hat{j})$ m/s. Its velocity (in m/s) at point B is: -

NEET-UG 2013 (+4 /-1)

1. $2\hat{i} + 3\hat{j}$
2. $-2\hat{i} - 3\hat{j}$
3. $-2\hat{i} + 3\hat{j}$
4. $2\hat{i} - 3\hat{j}$



Question 202

A projectile is fired from the surface of the earth with a velocity of 5 m/s and angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 m/s at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in m/s^2) is: (given $g = 9.8\ m/s^2$) **AIPMT 2014 (+4 /-1)**

1. 3.5
2. 5.9
3. 16.3
4. 110.8

$R_e = R_p$

$$\frac{u_e^2 \sin 2\theta}{g_e} = \frac{u_p^2 \sin 2\theta}{g_p}$$

$$\frac{25}{10} = \frac{3}{g_p} \qquad g_p = \frac{30}{25} = 3.5$$

Question 23

When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object the shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be

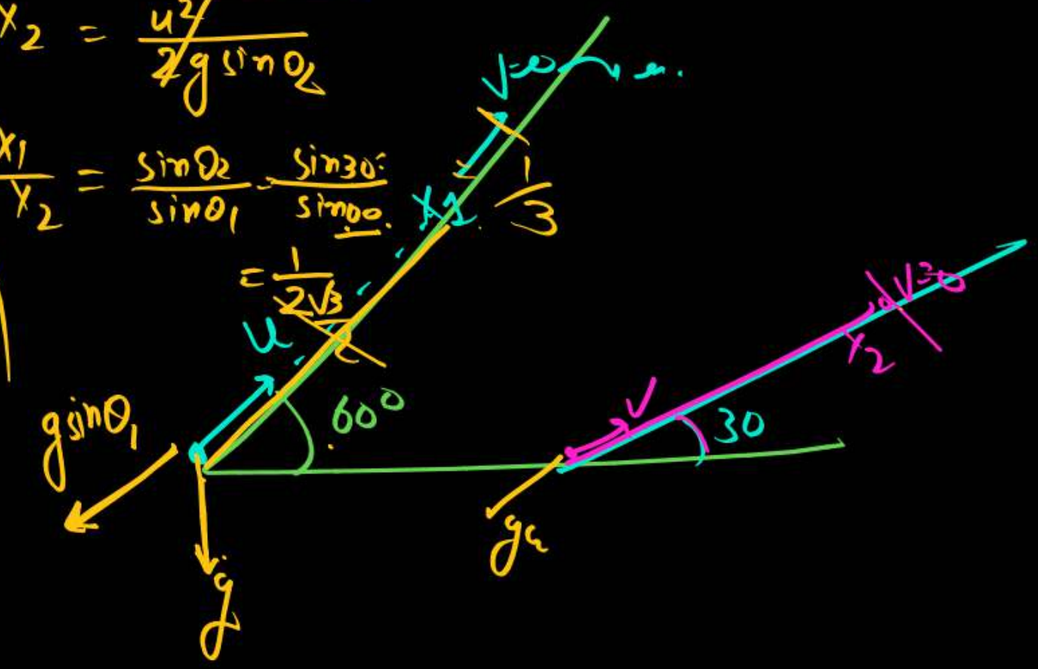
NEET(UG) 2019 (+4 /-1)

1. $1:\sqrt{2}$
2. $\sqrt{2}:1$
3. $1:\sqrt{3}$
4. $1:2\sqrt{3}$

$$x_1 = \frac{u^2}{2g\sin\theta_1}$$

$$x_2 = \frac{u^2}{2g\sin\theta_2}$$

$$\frac{x_1}{x_2} = \frac{\sin\theta_2}{\sin\theta_1} = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{1}{2\sqrt{3}}$$



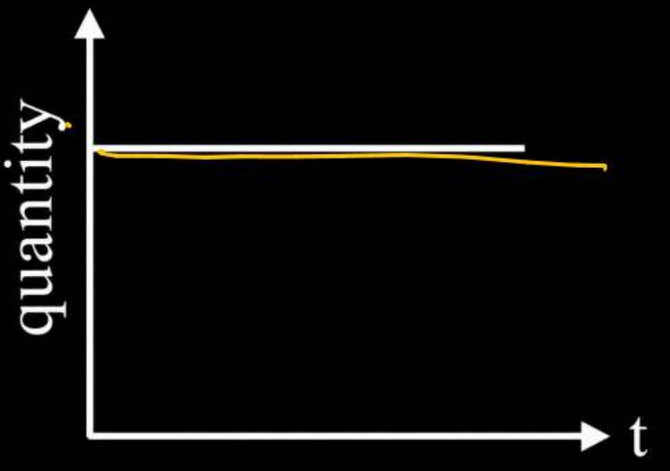
Projectile motion on Inclined

MR Scam

Question

224

In the graph shown in fig. time is plotted along x-axis. Which quantity associated with a projectile motion is plotted along the y - axis?



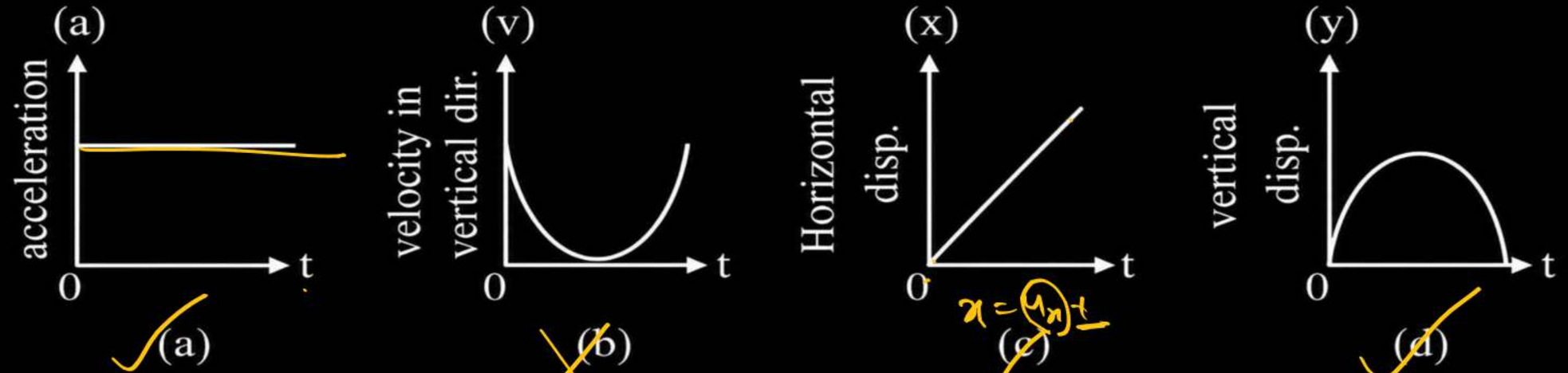
1. kinetic energy
2. momentum
3. horizontal velocity
4. none of the above

Question

25

For ground to ground projection following curves are given :-

Find incorrect one.



1. a
2. b
3. c
4. d

Question 24

In a projectile motion, the velocity :-

1. is always perpendicular to the acceleration
2. is never perpendicular to the acceleration
3. is perpendicular to the acceleration for one instant only
4. is perpendicular to the acceleration for two instants

Question 25

A body of mass m is thrown upwards at an angle θ with the horizontal with speed v. While rising up the magnitude of the velocity after t seconds will be:

1. $\sqrt{(v\cos\theta)^2 + (v\sin\theta)^2}$
2. $\sqrt{(v\cos\theta - v\sin\theta)^2 - gt}$
3. $\sqrt{v^2 + g^2t^2 - (2v\sin\theta)gt}$
4. $\sqrt{v^2 + g^2t^2 - (2v\cos\theta)gt}$

$\vec{V} = u\cos\theta\,\hat{i} + (u\sin\theta - gt)\,\hat{j}$

$|V| = \sqrt{u^2\cos^2\theta + (u\sin\theta - gt)^2}$

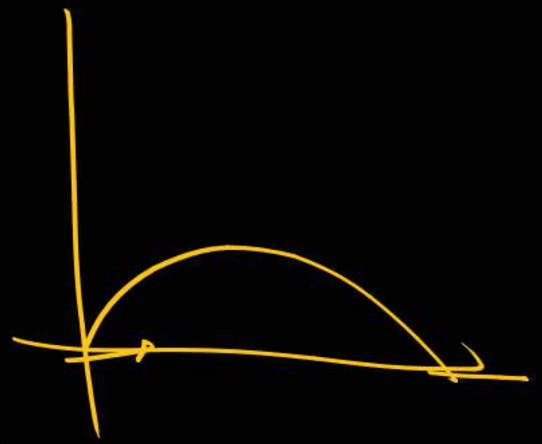
Solve by yourself

$= \sqrt{u^2 + g^2t^2 - 2u\sin\theta\, gt}$

Question 28

A particle is fired with velocity u making θ angle with the horizontal. What is the magnitude of change in velocity when it returns to the ground.

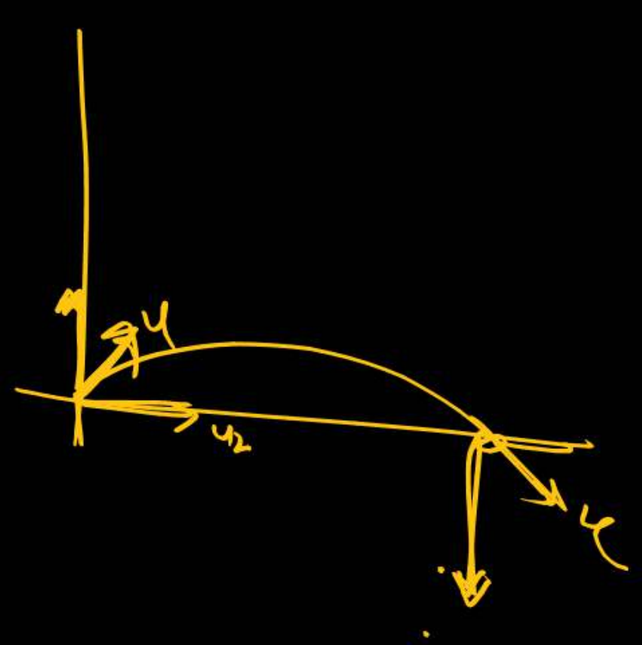
1. $u\cos\theta$
2. u
3. $2u\sin\theta$
4. $(u\cos\theta - u)$



Question 21

In the above, the change in speed is :

1. $u\cos\theta$
2. 0
3. $u\sin\theta$
4. $(u\cos\theta - u)$



Question 28

The angle which the velocity vector of a projectile, will make with the vertical after time t of its being thrown with a velocity $\mathbf{v}$ at an angle $\boldsymbol{\theta}$ to the horizontal, is :

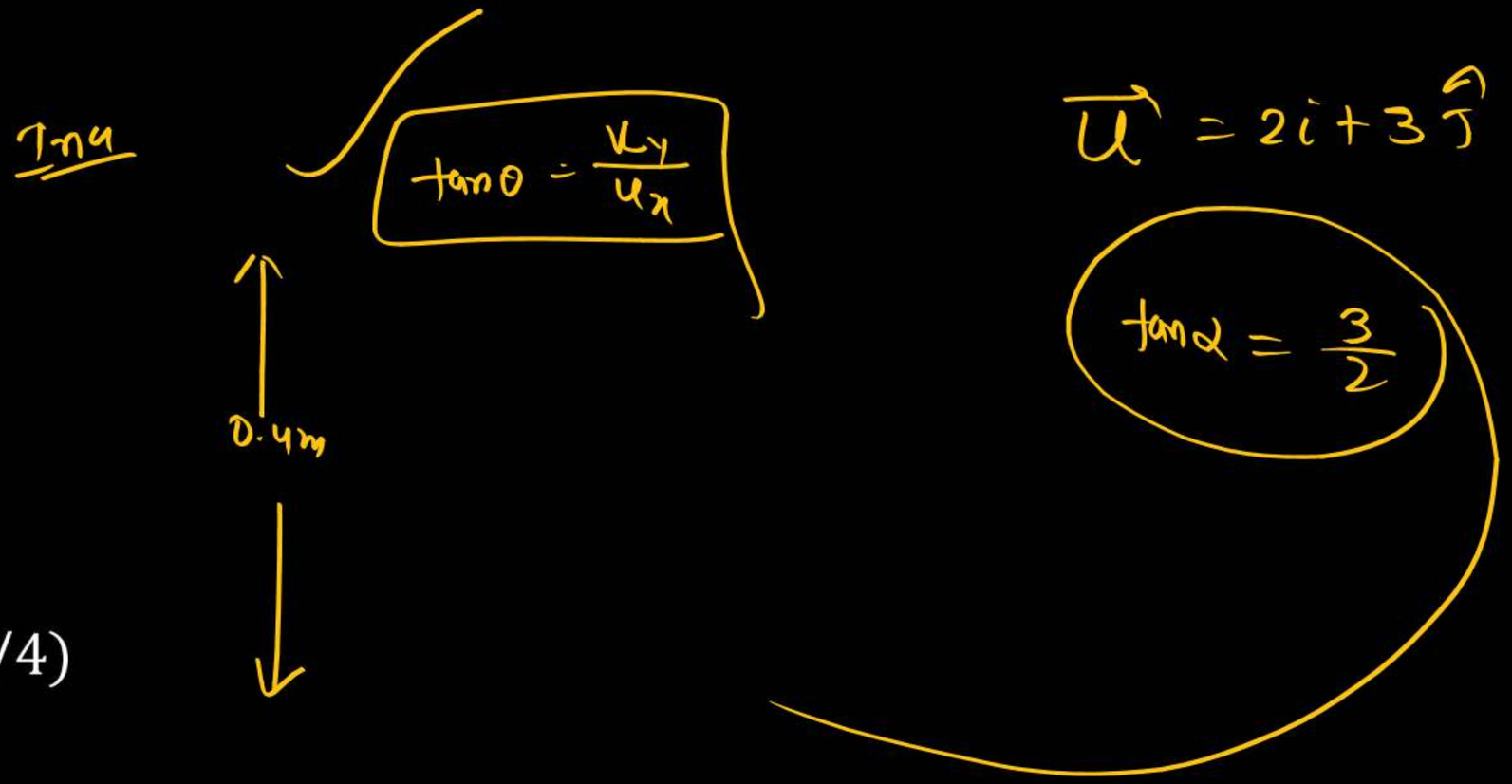
$\tan\alpha = \frac{v\sin\theta - gt}{v\cos\theta}$

1. θ
2. $\tan^{-1}\left(\frac{\theta}{t}\right)$
3. $\tan^{-1}\left(\frac{v\cos\theta}{v\sin\theta - gt}\right)$
4. ✓ $\tan^{-1}\left(\frac{v\sin\theta - gt}{v\cos\theta}\right)$

Question 31

At a height 0.4 m from the ground, the velocity of a projectile in vector form is : Z . 2j) m / s . The angle of projection from horizontal is $(\mathbf{g = 10\ m/s^2})$: –

1. 45°
2. 60°
3. 30°
4. $\tan^{-1}(3/4)$



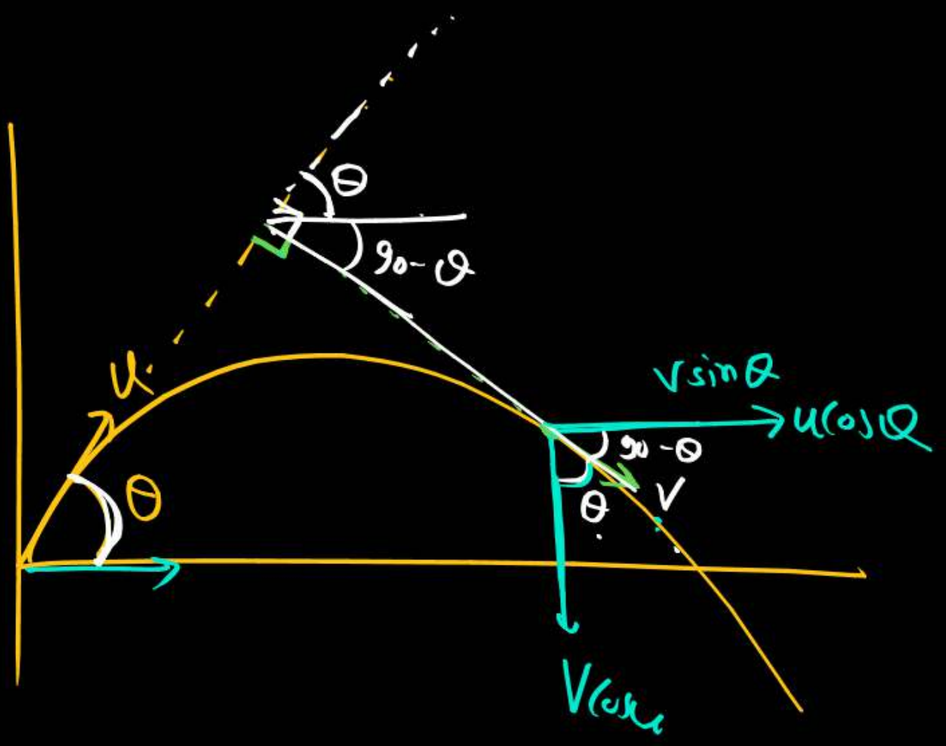
Question 39

Copy in note

{Try in rough}

A particle is projected with a velocity u making an angle θ with the horizontal. At any instant, its velocity v is at right angle to its initial velocity u; then v is:

1. $u \cos \theta$
2. $u \tan \theta$
3. $u \cot \theta$
4. $u \sec \theta$



$v \sin\theta = u \cos\theta$

$v = \frac{u\cos\theta}{\sin\theta} = u\cot\theta$

$v = u\cot\theta$

General 2-D Motion

Must do ael Question

Question

x and y-coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$, where x and y. Acceleration of the particle at $t = 2s$. **[NEET-2017]**

$\vec{a} = a_x \hat{i} + a_y \hat{j}$

$\vec{a} = -4\hat{i} + 0\hat{j}$

$\boxed{\vec{a} = -4\hat{i}}$

$x = 5t - 2t^2$

$V_x = \frac{dx}{dt} = 5 \times 1 - 2(2t)$

$V_x = (5 - 4t)\hat{i}$

$\frac{dV_x}{dx} = a_x = 0 - 4 \times 1$

$a_x = -4\hat{i}$

$y = 10t$

$\frac{dy}{dt} = V_y = 10 \times 1$

$V_y = 10\hat{j}$

$a_y = \frac{d10}{dt} = 0$

Question

A particle has initial velocity $2\hat{i} + 3\hat{j}$ and acceleration $(0.3\hat{i} + 0.2\hat{j})$. Magnitude of velocity after 10 sec. **[NEET-2012]**

$\vec{u}_{t=0} = 2i + 3\hat{j}$ $\quad \vec{a} = 0.3i + 0.2\hat{j}$

$u_x = 2$

$a_x = 0.3$

$\vec{v}_x = \vec{u}_x + \vec{a}_x t$

$\vec{v}_x = 2 + 0.3 \times 10$

$= 2 + 3$

$\vec{v}_x = 5i$

$\vec{v}_y = u_y + a_y t$

$= 3 + 0.2 \times 10$

$= 3 + 2$

$= 5\hat{j}$

$\vec{v} = v_x i + v_y \hat{j}$

$5 + 5 = 10$ ✗

$\vec{v} = 5i + 5\hat{j}$ ✓

$|v| = \sqrt{5^2 + 5^2}$

$= \sqrt{50} = 5\sqrt{2}$

Question

A position vector of a particle $r = 15t^2\,\hat{i} + (4 - 20t^2\,\hat{j})$. Find acceleration at $t = 1$ sec

[JEE Main 2019]

1. 25
2. 40
3. 100
4. 50

$\vec{r} = 15t^2\,\hat{i} + (4 - 20t^2\,\hat{j})$

$x = 15t^2$ $\qquad$ $y = 4 - 20t^2$

$\frac{dx}{dt} = 15(2t)$

$V_x = 30t$

$a_x = 30$

$V_y = \frac{dy}{dx} = 0 - 20(2t)$

$V_y = -40t$

$a_y = \frac{dV_y}{dt} = -40$

$\vec{a} = 30\hat{i} - 40\hat{j}$

$|a| = \sqrt{\quad}$

Question

Object starts from the point $(2\hat{i}, 4\hat{j})$ m at $t = 0$ with velocity $(5\hat{i} + 4\hat{j})$ with constant acceleration $(4\hat{i} + 4\hat{j})$ m/s^2. What is distance from particle from origin at t = 2 sec.

(x_f, y_f) **[JEE Main-2019]**

$(x_i, y_i) = (2i, 4j)$

$\vec{r} = x_f i + y_f j$

$u_x = 5i$	$u_y = 4j$
$a_x = 4$	$a_y = 4$
$s_x = u_x t + \frac{1}{2} a_x t^2$	$s_y = u_y t + \frac{1}{2} a_y t^2$
$x_f - x_i = 5 \times 2 + \frac{1}{2} \times 4 (2)^2$	$y_f - y_i =$
$x_f - 2 = 10 + 8$	
$x_f = 18 + 2 = 20$	

Question

Potion of object $\vec{r} = (t^2 - 38t)\,\hat{i} + 2t^3\,\hat{j}$ find instant when velocity and acceleration are perpendicular. **[JEE-2022]**

$$\vec{r} = (t^2 - 38t)\,\hat{i} + 2t^3\,\hat{j}$$

$$\frac{d\vec{r}}{dt} = (2t - 38)\,\hat{i} + 2(3t^2)\,\hat{j}$$

$$\vec{V} = (2t - 38)\,\hat{i} + 6t^2\,\hat{j}$$

$$\vec{a} = (2\times 1)\,\hat{i} + 6(2t)\,\hat{j}$$

$\vec{V} \perp \vec{a}$

$\vec{V} \cdot \vec{a} = 0$

$$4t - 76 + 72t^3 = 0$$

(a) 0

$t = 1$

$$4t - 76 + 72t^3 = 0$$

$$4\times 1 - 76 + 72(1)^3 = 0$$

$$4 - 76 + 72 = 0$$

Question

A particle starts from origin with velocity $3\hat{j}$ m/s and acceleration $(6\hat{i} + 4\hat{j})$. Then find x-coordinates of particle when y-coordinates is 32. **[IIT-2021]**

$(x_i, y_i) = (0,0)$

$\vec{u} = 0\hat{i} + 3\hat{j}$

$\vec{a} = 6\hat{i} + 4\hat{j}$

$x_f = ?$

$x_f - x_i = u_x t + \frac{1}{2} a_x t^2$

$x_f = \frac{1}{2} 6 (t)^2$

$\boxed{x_f = 3(t)^2}$

$y_f = 32$

$y_f - 0 = 3t + \frac{1}{2} 4(t)^2$

$32 = 3t + 2t^2$

$2t^2 + 3t - 32 = 0$

$t = \frac{-3 + \sqrt{9 - 4 \times 2(-32)}}{2 \times 2}$

$t = \frac{-3 + \sqrt{9 + 8 \times 32}}{4}$

$t = \frac{-3 + \sqrt{265}}{4}$

$t \approx 16$

$\frac{32}{8}$

$\frac{256}{265}$

Question

H/W

Object is moving with velocity $\vec{V} = 3 \sin(\omega t)\,\hat{i} + 3 \cos(\omega t)\,\hat{j}$ then find distance moved by object in 2 sec.

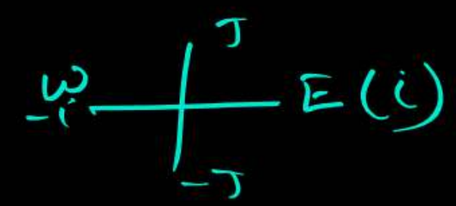
Question

Object is moving in west with 5 m/s after 2 sec its velocity is 5 m/s in north then find acceleration.

NEET

$\vec{u}_i = -5 m/s\, \hat{i}$

$V_f = 5 m\, \hat{j}$



dir of accn alg force ✓
or alg change in ve
$\Delta v = \vec{V_f} - \vec{V_i}$

$$\vec{a} = \frac{\vec{V_f} - \vec{V_i}}{t}$$

$$= \frac{5\hat{j} - (-5\hat{i})}{2}$$

$$\vec{a} = \frac{5\hat{j} + 5\hat{i}}{2} = \frac{5}{2}(\hat{i} + \hat{j})$$ (North east)

gn

Question 32

Ball is projected with 30 m/s in horizontal direction from some height. Find time when it is 45° from horizontal.

Question 31

A ball is projected upwards from the top of a tower with a velocity of 50 m/s making an angle of 30° with the horizontal. The height of the tower is 70 m. After how much time from the instant of throwing, will the ball reach the ground?

1. 2 s
2. 5 s
3. 7 s
4. 9 s

Question

When a particle is thrown horizontally, with initial velocity 'u' the resultant velocity of the projectile at any time t is given by:

1. gt
2. $\frac{1}{2}gt^2$
3. $\sqrt{u^2 + g^2t^2}$
4. $\sqrt{u^2 - g^2t^2}$

Question

Ball is projected with 30 m/s in horizontal direction from some height. Find time when it is 45° from horizontal.

Question

A body is thrown horizontally from the top of a tower of height 5 m. It touches the ground at a distance of 10 m from the foot of the tower. The initial velocity of the body is (g = 10 ms)

1. 2.5 ms
2. 5 ms
3. 10 ms
4. 20 ms

Question

A body is thrown horizontally with a velocity $\sqrt{2gh}$ from the top of a tower of height h. It strikes the level ground through the foot of the tower at a distance x from the tower. The value of x is:

1. h
2. $h/2$
3. $2h$
4. $2h/3$

Question

A bomber is flying horizontally with a constant speed of 150 m/s at a height of 78.4 m. The pilot has to drop a bomb at the enemy target. At what horizontal distance from the target should he release the bomb:

1. Zero
2. 300 m
3. 600 m
4. 750 m

Question

A car starts from rest and accelerates at 5 m/s^2. At t = 4 s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at = 6 s ?

(Take g = 10 m/s^2)

NEET(UG) 2021 (+4 /-1)

1. 20 m/s, 5 m/s^2
2. 20 m/s, 0
3. $20\sqrt{2}$ m/s, 0
4. $20\sqrt{2}$ m/s, 10 m/s^2

Question

Two bullets are fired horizontally and simultaneously towards each other from roof tops of two buildings 100 m apart and of same height of 200m with the same velocity of 25 m/s. When and where will the two bullets collide. ($g = 10\ m/s^2$)

NEET(UG) 2019 (Odisha) (+4 /-1)

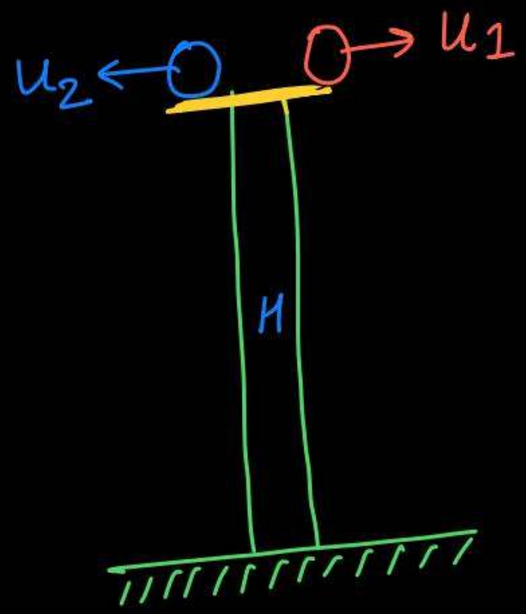
1. after 2s at a height 180 m
2. after 2s at a height of 20 m
3. after 4 s at a height of 120 m
4. they will not collide

Question

A ball is projected upwards from the top of a tower with a velocity of 50 m/s making an angle of 30° with the horizontal. The height of the tower is 70 m. After how much time from the instant of throwing, will the ball reach the ground?

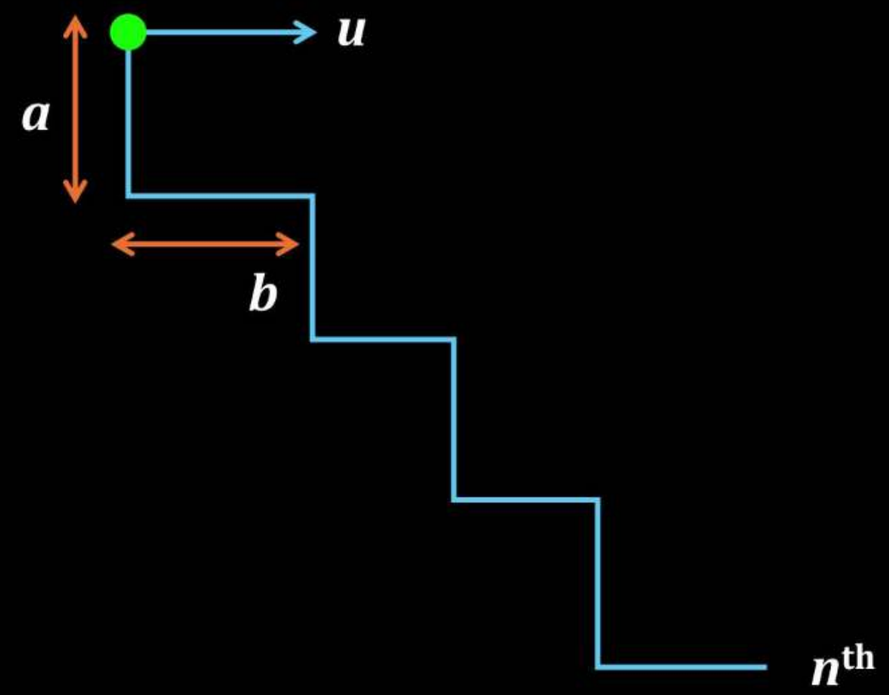
1. 2 s
2. 5 s
3. 7 s
4. 9 s

Two Ball Projected Horizontaly from Height H. in opposite direction with speed u_1 and u_2 then find time when they moving perpendicular to each other and also find Horizontal distance between them when they are moving perpendicular



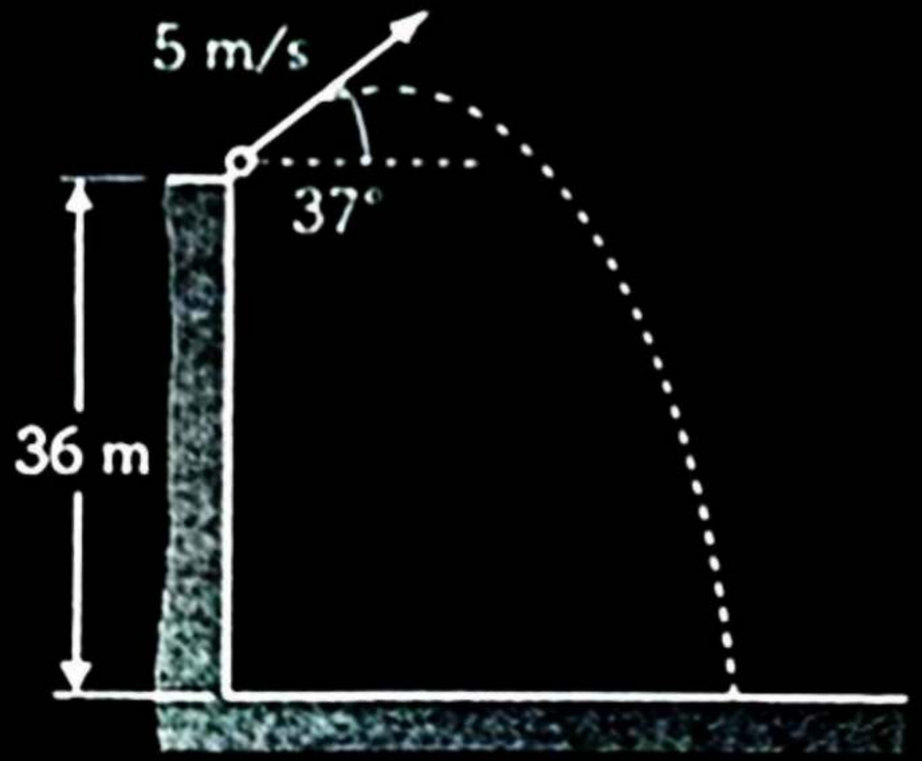
Question

Find velocity so that ball will fall on n^{th} step.



Question

A ball is thrown from the top of 36 m high tower with velocity 5 m/s at an angle 37° above the horizontal as shown. Its horizontal distance on the ground is closest to [$g = 10\ m/s^2$]



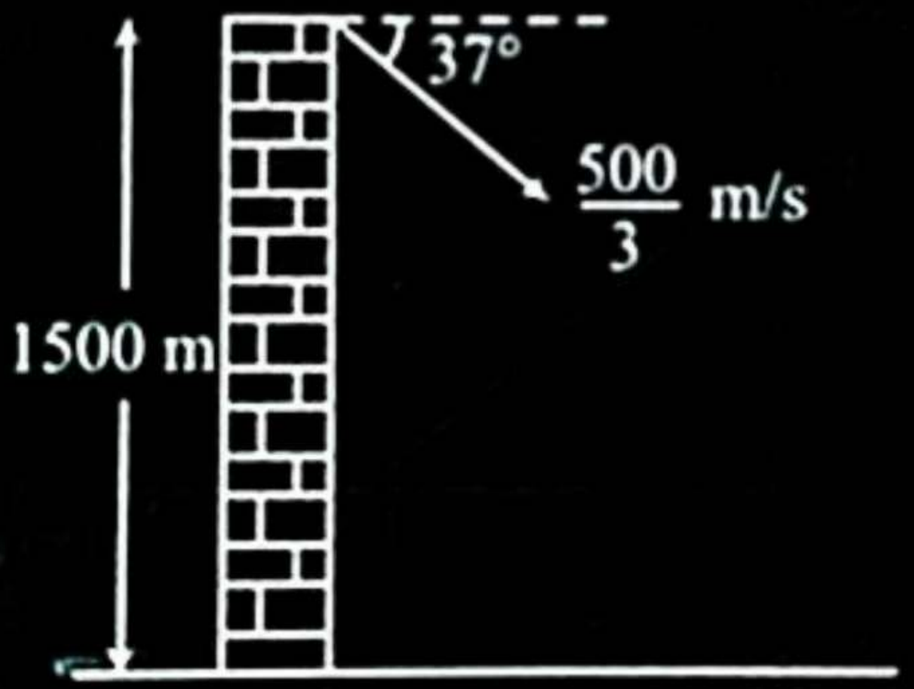
1. 12 m
2. 18 m
3. 24 m
4. 30 m

Question

A particle is projected from a tower as shown in figure, then the distance from the foot of the tower where it will strike the ground will be:

(take $g = 10\ m/s^2$)

1. 4000/3 m
2. 5000/3 m
3. 2000 m
4. 3000 m



Thank You